

INDIANA COUNTY AP, PA, USA

WMO: 725175

Lat: 40.633N

Lon: 79.100W

Elev: 1405

StdP: 13.97

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 64706

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.7	8.5	-6.1	4.2	9.0	0.1	5.8	11.3	24.8	32.2	22.0	29.9	5.8	260	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	88.0	73.4	84.5	71.6	82.3	70.3	76.3	83.3	74.9	81.3	73.5	79.0	7.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	136.1	79.5	72.9	128.5	77.4	72.2	125.4	76.8	40.8	83.3	39.4	82.3	38.1	77.7	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	17.8	15.9	-5.8	90.6	4.4	3.5	-9.0	93.2	-11.6	95.2	-14.0	97.2	-17.2	99.8		
			-6.1	78.6	4.6	2.2	-9.4	80.2	-12.1	81.5	-14.7	82.8	-18.0	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	27.1	30.3	38.7	50.7	60.7	68.0	71.8	69.8	64.7	53.0	42.7	32.7	
	DBStd	17.96	12.19	11.77	10.92	9.62	8.32	6.14	5.02	4.85	7.00	8.90	9.26	9.84	
	HDD50	2640	713	561	382	105	12	0	0	0	1	70	256	540	
	HDD65	5786	1173	972	815	437	183	40	7	14	90	385	670	1000	
	CDD50	2994	5	9	32	127	344	539	677	615	441	164	36	5	
	CDD65	667	0	0	0	10	50	129	219	164	81	14	0	0	
	CDH74	4995	0	0	4	108	454	994	1644	1104	600	83	4	0	
	CDH80	1258	0	0	0	17	90	269	478	239	154	11	0	0	
Wind	WSAvg	5.9	7.9	7.6	7.2	7.0	5.3	4.7	4.2	3.7	4.0	5.3	6.5	7.5	
Precipitation	PrecAvg	46.7	4.1	3.0	4.1	4.2	4.0	4.5	5.0	4.2	3.8	3.1	3.7	3.8	
	PrecMax	58.5	8.5	5.4	7.3	7.0	6.3	7.0	9.1	8.6	8.2	6.1	8.0	7.4	
	PrecMin	36.8	1.9	1.9	1.9	2.0	1.7	1.9	2.6	1.8	1.3	1.3	1.0	2.3	
	PrecStd	5.3	1.6	0.9	1.3	1.3	1.3	1.4	1.6	2.0	1.8	1.3	1.6	1.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	66.5	73.0	82.1	86.3	89.8	90.9	88.4	88.4	81.4	72.0	63.3	
		MCWB	55.7	56.7	58.8	65.2	70.5	72.5	74.8	73.1	71.5	70.8	61.5	57.5	
	2%	DB	55.3	59.2	67.1	77.0	82.0	86.2	88.1	84.4	84.1	75.2	66.0	57.0	
		MCWB	51.6	52.6	55.9	60.6	67.6	72.3	74.2	72.1	71.1	65.0	58.8	52.6	
	5%	DB	50.2	54.1	61.4	72.3	79.5	83.5	84.4	82.1	81.1	72.2	62.9	53.6	
		MCWB	46.5	48.3	51.9	58.1	65.8	70.7	72.3	70.8	69.6	63.5	55.4	50.3	
	10%	DB	45.1	47.6	55.5	67.5	75.1	80.7	82.0	80.8	77.0	67.9	58.6	47.6	
		MCWB	41.9	42.9	48.4	55.8	64.2	69.4	71.0	70.1	67.6	61.9	52.5	44.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.3	58.3	61.9	68.4	74.2	76.7	79.3	77.6	77.0	72.8	64.6	59.0
			MCDB	60.1	63.3	68.5	78.0	82.4	84.4	86.7	83.9	82.9	78.7	67.9	62.0
2%		WB	53.8	54.3	58.1	64.5	70.9	74.9	76.9	75.4	74.3	69.3	60.5	54.4	
		MCDB	57.0	58.1	63.7	71.1	78.4	82.3	84.2	81.6	79.6	73.2	64.6	56.6	
5%		WB	46.6	48.5	54.7	61.6	68.7	73.2	75.0	73.9	72.2	65.9	56.8	49.9	
		MCDB	48.1	53.3	60.3	68.1	75.5	79.8	81.4	78.8	77.0	69.7	60.8	52.5	
10%		WB	41.5	43.0	50.0	58.5	66.7	71.5	73.5	72.4	70.1	62.7	53.4	44.6	
		MCDB	44.8	47.4	56.1	65.1	72.9	77.3	79.1	76.8	74.5	66.3	57.8	46.9	
Mean Daily Temperature Range	5% DB	MDBR	14.7	16.4	18.1	21.3	20.8	19.2	18.9	18.6	19.2	18.1	16.5	13.6	
		MCDBR	18.5	22.9	25.1	27.6	24.6	22.4	21.4	21.0	23.1	23.5	22.6	20.9	
		MCWBR	15.4	17.6	16.1	14.5	12.1	10.4	9.4	9.8	11.0	13.3	15.4	18.1	
	5% WB	MCDBR	17.1	21.7	21.9	22.3	20.4	18.5	17.9	16.0	17.3	19.1	19.5	19.7	
		MCWBR	15.8	18.2	16.2	14.1	11.4	10.2	9.3	8.8	9.5	12.4	15.8	18.4	
Clear-Sky Solar Irradiance	taub	0.312	0.328	0.354	0.378	0.418	0.459	0.470	0.462	0.381	0.366	0.322	0.303		
	taud	2.404	2.339	2.355	2.327	2.229	2.144	2.129	2.139	2.348	2.386	2.495	2.484		
	Ebn at Noon	268	279	284	284	273	261	257	255	271	262	263	263		
	Edh at Noon	26	32	36	39	44	48	48	47	35	30	24	23		
All-Sky Solar Radiation	RadAvg	482	743	1050	1428	1636	1789	1808	1622	1322	887	601	397		
	RadStd	42	97	103	138	151	151	119	97	148	93	75	42		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+95	

Nomenclature: See separate page